Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI-based Expert System and LLM-driven Automatic CNC Program Generation for Grinding Processes			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical·Materials			
	Medium Category	Small Category	Medium Category	Small Category
	Precision manufacturing machinery	Cutting Machine Tools, Grinding/Polishing Machine Tools, Numerical Control (NC) Units, CAD/CAM-related Software		
Concept of Proposed Technology	The proposed technology is a grinding-specific, AI-assisted software framework for safe and explainable automatic CNC program generation and process planning in precision grinding. Users provide a compact set of application-oriented inputs, including (i) part definition via parameterized geometry, CAD data, or 2D technical drawings, (ii) workpiece material state (material and hardness), (iii) grinding wheel and dressing tool constraints (e.g., tool size and availability), and (iv) target quality requirements such as tolerances and surface roughness. A feature interpretation module transforms the provided geometry and quality information into a structured, manufacturing-relevant representation, capturing grinding-critical features (e.g., functional surfaces, transitions, tolerance zones, and surface finish constraints) required for process planning. This interpretation includes AI-supported semantic analysis of 2D technical drawings (e.g., segmentation of geometric entities and automated interpretation of dimensional information) to enable reliable extraction of grinding-relevant features. Based on this structured representation, a knowledge-based expert system encapsulating validated grinding process knowledge selects feasible dressing and grinding strategies, tool combinations, and parameter windows [1, 2]. The expert			

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	system generates a machine-independent process plan with explicit constraints and traceable decision logic, ensuring deterministic and explainable process planning. A constrained, AI-assisted large language model (LLM) is subsequently employed to translate the expert system outputs into controller-specific CNC programs (e.g., Siemens SINUMERIK), operating strictly within predefined syntax rules, parameter limits, and controller logic [3]. Deterministic validation checks ensure rule compliance, completeness, and controller compatibility. Integrated troubleshooting and process risk assessment functions evaluate the planned operating point against known grinding defect mechanisms, such as chatter, thermal damage, and form deviations, and propose corrective parameter or strategy adjustments within the validated process envelope. A human-in-the-loop workflow ensures industrial safety and acceptance, requiring the operator to review an explainable summary of selected strategies, constraints, and identified risks before approving the CNC program for execution.
Rationale for Joint R&D Project	Manufacturing industries are facing a persistent and structural shortage of highly qualified CNC programmers, a challenge that is particularly critical in precision grinding operations. In grinding, CNC programming requires not only syntactic knowledge of controller commands, but also deep process understanding of wheel–workpiece interaction, dressing strategies, thermal behavior, and quality assurance [4]. Such expertise is largely experience-based, tacit, and strongly operator-dependent, which makes it difficult to scale across products, machines, and production sites—especially for small and medium-sized enterprises (SMEs). Recent research at the proposing institute has shown that knowledge-based expert systems and adaptive decision-making frameworks can successfully formalize grinding know-how into deterministic rules, parameter windows, and quality-driven constraints, significantly reducing reliance on individual expert programmers while maintaining process robustness and product quality [2], [5]. These approaches establish a reliable foundation for systematic process planning, troubleshooting, and explainable decision support in an Industry 4.0 context. However, a critical gap remains between validated process planning results and their efficient implementation on industrial CNC controllers. Existing automated CNC program generation solutions lack grinding-specific process intelligence, explicit safety constraints and transparent decision-making logic. This limits their applicability to high-precision and safety-critical grinding operations. [6]. To address this gap, the proposed project introduces a constrained, AI-assisted large language model (LLM) as a controlled code-generation component. The LLM is not used for autonomous decision-making, but strictly for the automatic generation of controller-specific CNC programs (e.g., Siemens SINUMERIK) based on expert-system-validated process plans, predefined syntax templates, parameter limits, and safety rules. This approach enables efficient and scalable CNC code generation while preserving deterministic behavior, full traceability, and operator oversight through a human-in-the-loop workflow. Practical industrial deployment of such a system requires robust software architectures, seamless integration with CNC controllers, secure data handling, and operator-oriented interfaces compatible with existing production environments. This is where Korean SMEs and startups contribute decisive strengths, including expertise in CNC systems, industrial software platforms, digital manufacturing infrastructures, and scalable productization. Joint R&D is therefore essential to combine deep grinding process intelligence with industrial-grade software engineering and CNC integration. By uniting the proposing institute's validated expert-system-based grinding knowledge with the Korean partners' capabilities in CNC integration and digital platforms, the project enables the development of a deployable, safe, and explainable system for LLM-assisted automatic CNC program generation in precision grinding. The joint approach directly supports TIPA's objectives by accelerating technology transfer from research to SME-level production, reducing implementation risk, and delivering a commercialization-ready solution that addresses workforce shortages, productivity demands, and quality consistency in advanced manufacturing.
Key Words	1. Automatic CNC Program Generation 2. Knowledge-Based Expert Systems 3. Large Language Model (LLM) 4. AI-Assisted and Constrained Artificial Intelligence 5. Precision Grinding Processes

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (7)
R&D Objectives	• Development of a configurable, knowledge-based expert system for precision grinding, adaptable to different grinding processes and machine configurations. • Automatic selection of dressing tools, grinding wheels, dressing and grinding strategies, and parameter windows based on part geometry and quality requirements. • AI-supported feature interpretation from CAD models and 2D technical drawings, including semantic analysis of geometry and dimensional information, to enable grinding-specific process planning. • Integration of troubleshooting and corrective decision logic for common grinding defects (e.g., chatter, thermal damage, form deviations). • LLM-assisted, controller-specific CNC program generation with deterministic constraints and traceable decision paths. • Industrial integration into CNC systems via a secure, scalable, and operator-oriented software platform.			
R&D Target Specifications	• Reduction of CNC programming time by more than 70% compared to conventional manual programming for representative precision grinding tasks. • Reduction of scrap risk by more than 80% through expert-system-based selection of dressing and grinding tools and strategies, and validated parameter windows, combined with deterministic CNC program generation. • Reliable achievement of specified dimensional tolerances and surface roughness requirements (e.g., Ra ≤ 0.4 µm), supported by automated planning and confirmed through industrial validation tests. • Support for industrial CNC controllers, initially Siemens SINUMERIK, with a modular architecture extensible to additional controller platforms. • Deterministic, verifiable, and explainable CNC code generation, including LLM-assisted translation under strict constraints, with full traceability of decision logic, parameter selection, and applied safety limits.			
R&D Plan	The proposed R&D activities are structured based on a three-year project framework, which provides sufficient time for systematic development, industrial integration, and validation of the proposed edge-based AI solution. The detailed timeline, task sequencing, and milestone allocation will be jointly defined and finalized in close coordination with the Korean industry partner prior to the project start, taking into account the partner's technical focus, available infrastructure, and commercialization strategy. All work packages, task contents, assignment of work-package leads and responsibilities, and technical specifications described are to be understood as a conceptual framework and will be closely coordinated with the Korean development partner, adapted as necessary, and further developed jointly to ensure optimal technical implementation and high industrial relevance. The final R&D plan will be discussed, adjusted, and mutually agreed upon with the Korean SME or startup partner to ensure optimal efficiency, clear ownership, and strong industrial relevance throughout the project duration. The final R&D plan will be discussed, adjusted, and mutually agreed upon with the Korean SME or startup partner to ensure optimal efficiency, clear ownership, and strong industrial relevance throughout the project duration. The overall methodology combines four tightly coupled components: i. Formalized domain expert knowledge in grinding technology, implemented through data-driven and rule-based expert systems; ii. AI-supported feature interpretation of geometric and quality-related information extracted from CAD models and 2D technical drawings; ii. Constrained, AI-assisted CNC code generation using large language models (LLMs) operating under strict syntactic, semantic, and safety constraints; and v. Industrial-grade CNC system integration, including controller interfaces, validation workflows, and operator-oriented user interfaces. Safe automation with human-in-the-loop validation is the core principle. Grinding process knowledge is first formalized into deterministic expert rules, quality-driven constraints, and validated parameter windows. These machine-independent			

process plans constitute the sole input to CNC code generation. LLMs are used exclusively as controlled translation tools to generate controller-specific CNC programs (e.g., Siemens SINUMERIK) based on predefined templates, parameter limits, and controller logic. Autonomous decision-making by the LLM is explicitly excluded. All generated CNC programs undergo deterministic validation and require explicit operator approval prior to execution.
The system is designed to be configurable and extensible, enabling adaptation to different grinding processes (e.g., cylindrical plunge, traverse, surface grinding), machine configurations, and CNC controllers..

Work Packages, Tasks, and Methodologies:

WP1 – Grinding Knowledge Formalization and Expert System Development
Lead (Optional-Suggestion): Research Institute
Objectives:
Develop a validated knowledge-based expert system for deterministic grinding process planning based on part geometry, material properties, and quality requirements.
Key Tasks:

- Identification and classification of representative grinding applications (e.g., cylindrical plunge grinding, traverse grinding, surface grinding)
- Systematic formalization of grinding process knowledge, including:
  a. Selection of abrasive type and grain size
  b. Bond system selection
  c. Wheel structure and abrasive concentration
  d. Dressing tools and dressing strategies
  e. Validated parameter windows (cutting speed, feed rate, depth of cut, dressing parameters)
- Definition of quality-driven constraints, including surface roughness targets, tolerance classes, and process stability limits
- Encoding of grinding knowledge using rule-based and hybrid (rule + data) expert system approaches

Methodology:

- Expert interviews and structured knowledge acquisition
- Literature-based consolidation of grinding process models
- Rule-based expert system design with traceable decision logic
- Validation using reference grinding experiments and historical process data

Milestone M1:
Validated expert system for tool, strategy, and parameter selection.

WP2 – Feature Interpretation from Geometry and Quality Requirements
Lead (Optional-Suggestion): Joint (Research Institute + Korean Partner)
Objectives:
Develop a robust feature interpretation module that converts CAD data and 2D technical drawings into a structured, manufacturing-relevant representation for grinding-specific process planning.
Key Tasks:

- Definition of grinding-relevant geometric and functional features.
- Interpretation of dimensional tolerances and surface roughness specifications.
- AI-supported semantic interpretation of geometric entities and dimensional information in technical drawings.
- Mapping of extracted features to grinding strategies, process constraints, and expert-system inputs.
- Development of standardized interfaces for CAD model and drawing uploads.

Methodology:

- Feature-based design and manufacturing principles
- Rule-based and AI-supported geometry interpretation methods
- Structured data models for part geometry, quality attributes, and manufacturing intent

Milestone M2:
Validated feature interpretation module providing structured, grinding-specific inputs to the expert system.

WP3 – LLM-Based CNC Program Generation for Grinding
Lead (Optional-Suggestion): Joint (Research Institute + Korean Partner)
Objectives:
Translate expert-system-validated process plans into safe, controller-specific CNC programs.
Key Tasks:

- Definition of structured interfaces between the expert system outputs and the LLM input layer.
- Development of controller-specific CNC templates and syntax rules (initial focus: Siemens SINUMERIK).
- Implementation of constrained LLM prompting for grinding and dressing cycles.

- Integration of deterministic syntax checks, rule compliance validation, and controller compatibility verification.

Methodology:

- Prompt engineering with strict output constraints
- Controller-specific post-processing and validation pipelines

Milestone M3:

Automatic generation of syntactically correct, controller-compatible, and verifiable CNC grinding programs.

WP4 – Troubleshooting and Process Risk Assessment

Lead (Optional-Suggestion): Research Institute

Objectives:

Integrate explainable troubleshooting and process risk assessment capabilities into the automated process planning workflow.

Key Tasks:

- Identification and classification of common grinding defects (e.g., chatter, thermal damage, form deviations).
- Modeling of cause–effect relationships.
- Development of rule-based diagnostic and decision logic.

Methodology:

- Cause–effect modeling of grinding defects
- Expert-rule-based diagnostic frameworks combined with empirical process limits
- Decision-tree-based troubleshooting logic

Milestone M4:

Integrated troubleshooting module providing explainable risk assessment and corrective actions.

WP5 – Industrial Platform and CNC Integration

Lead (Optional-Suggestion): Korean SME / Startup

Objectives:

Deploy the developed expert system and AI-assisted CNC generation workflow on an industrial-grade software platform connected to CNC grinding machines.

Key Tasks:

- Design and implementation of system architecture and operator-oriented user interfaces
- Integration with CNC controllers via industrial communication standards (e.g., OPC UA)
- Implementation of data security, access control, and system reliability mechanisms
- Visualization of generated CNC programs, decision logic, and troubleshooting results

Methodology:

- Industrial software engineering practices
- CNC controller APIs and OPC UA integration
- User-centered interface design

Milestone M5:

Fully integrated prototype system connected to an industrial grinding machine.

WP6 – Validation, Demonstration, and Pilot Operation

Lead (Optional-Suggestion): Joint

Objectives:

Validate system performance, robustness, and usability under real industrial conditions.

Key Tasks:

- Pilot testing on industrial cylindrical grinding machines
- Quantitative evaluation of CNC programming time reduction
- Operator feedback and iterative refinement

Milestone M6:

TRL 7 system validated in an industrial environment.

Work Packages Overview (36 Months):

- WP1 – Expert system core
- WP2 – Feature interpretation
- WP3 – LLM-based CNC generation
- WP4 – Troubleshooting
- WP5 – Industrial integration
- WP6 – Validation and pilot operation

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	The research institute will serve as the scientific lead and primary provider of grinding process intelligence, ensuring that all automation and AI-assisted components are grounded in validated process knowledge, deterministic logic, and industrial safety principles. Its role is to translate long-standing grinding expertise into a formalized, explainable, and transferable digital framework suitable for reliable industrial deployment. Key responsibilities include: • Formalization of grinding process knowledge into a structured, knowledge-based expert system, covering grinding strategy selection, dressing logic, tool–workpiece interaction constraints, and process parameter windows. • Development of process- and physics-aware grinding models linking part geometry, material properties, dressing and grinding conditions, and operating parameters to achievable surface quality, dimensional accuracy, and process stability. • Definition of quality-driven and safety-critical constraints, including surface roughness limits, tolerance classes, thermal damage thresholds, and chatter avoidance regions. • Design of the hybrid decision-making architecture, combining deterministic expert rules with AI-assisted automation while maintaining full traceability and explainability of all decisions. • Specification of LLM usage boundaries, including structured inputs, constrained outputs, controller-aware syntax rules, and deterministic validation mechanisms, ensuring that the LLM functions strictly as a controlled code-generation assistant rather than a black-box decision maker. • Development of integrated troubleshooting and process risk assessment logic, systematically linking known grinding defect mechanisms (e.g., burn, chatter, form deviations) to root causes and corrective actions within validated process envelopes. • Scientific validation of the developed methods through controlled laboratory experiments, benchmark grinding cases, and robustness analyses, ensuring reproducibility and transferability across machines and applications. • Joint execution and supervision of pilot operation phases, including interpretation of industrial test results and refinement of process models based on real production feedback. • Dissemination of project results through publications, patent applications, and technology transfer documentation, supporting long-term innovation, standardization, and future research collaboration.
Expected R&R of Korean SME Partner	The Korean SME or startup partner will lead the industrialization, system integration, and commercialization of the developed technology, ensuring that research outcomes are transformed into a robust, deployable, and scalable industrial solution aligned with real SME production environments. Key responsibilities include: • Design and implementation of an industrial-grade digital software platform enabling automated grinding process planning and CNC program generation in real production environments. • Development and maintenance of CNC controller interfaces, initially targeting Siemens SINUMERIK, with a modular architecture that allows extension to additional industrial controllers and machine configurations. • Integration of the expert system, feature extraction modules, troubleshooting logic, and AI-assisted CNC generation components into machine tool environments, ensuring real-time compatibility and operational stability. • Development of operator-oriented human–machine interfaces (HMIs) that present generated CNC programs, selected strategies, constraints, and troubleshooting recommendations in a transparent and usable manner. • Implementation of data management, cybersecurity, access control, and system reliability mechanisms, aligned with industrial IT and OT standards and SME operational requirements. • Execution of industrial pilot tests in SME production environments, including structured collection of performance metrics, usability feedback, and system reliability data. • Iterative refinement of the platform based on operator feedback, production constraints, and integration challenges encountered during pilot operation. • Contribution to business development strategies, post-project deployment planning, and market scaling within the Korean and international manufacturing ecosystem.

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The project will deliver a validated system for safe, explainable, and automated CNC program generation in precision grinding, demonstrated under real industrial conditions. The developed solution will enable a fundamental shift from manual, experience-driven CNC programming toward knowledge-based, AI-assisted, and human-supervised process planning. Technical and Industrial Outcomes: • A deployable software system that reduces CNC programming time by more than 70%, while maintaining deterministic, verifiable, and controller-compatible CNC code generation. • Significant reduction of scrap rates and rework through systematic selection of grinding strategies, dressing conditions, and process parameter windows, directly improving process stability and product quality. • Improved process consistency and reproducibility across machines, operators, and production sites, particularly in high-precision grinding applications with tight tolerances and surface integrity requirements. • Integration of explainable troubleshooting and process risk assessment, enabling early identification and mitigation of defects such as chatter, thermal damage, and form deviations. Impact on Korean SMEs and Commercialization Potential: • Korean SMEs and startups will obtain ownership of an industrial-grade digital manufacturing platform that can be customized and scaled across different grinding machines, customers, and application domains. • The solution supports commercial differentiation by embedding advanced grinding process intelligence into software products, rather than relying solely on machine hardware or operator expertise. • Reduced dependency on scarce expert programmers enables SMEs to scale production capacity and product variety without proportional increases in skilled labor. • The modular platform architecture facilitates future extensions to additional machining processes, CNC controllers, and digital manufacturing services. Workforce and Societal Impact: • The system addresses the growing shortage of highly skilled CNC programmers by transferring tacit grinding expertise into a digital, reusable, and explainable form. • Operators are supported through human-in-the-loop workflows, maintaining control and responsibility while benefiting from advanced decision support. • The approach contributes to upskilling and knowledge preservation, ensuring that critical manufacturing know-how remains available despite demographic change and workforce turnover. Strategic and Long-Term Impact: • The project strengthens international research–industry collaboration between Germany and Korea, accelerating technology transfer from advanced manufacturing research to SME-level production. • It contributes to smart manufacturing and AI-driven production systems that prioritize safety, explainability, and industrial acceptance. • The developed technology establishes a foundation for future standardization, certification, and cross-sector adoption of AI-assisted CNC programming solutions in precision manufacturing.

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization